Calculus Review, Part I

Bridging Mathematics, Fall 2025

1. Consider the limit
$$\lim_{x\to k} \frac{x^2-4x+4}{x^3+5x^2-14x}.$$
Does the limit exist when $k=0$? What about when $k=2$?

To begin, observe that
$$\frac{x^2-4x+4}{x^3+5x^2-14x} = \frac{(x-2)(x-2)}{x(x^2+5x-14)} = \frac{(x-2)^2}{x(x-2)(x+7)} = \frac{x-2}{x(x+7)}.$$
When $k=0$, this limit does not exist, in particular,
$$\lim_{x\to 0^-} \frac{x-2}{x(x+7)} = \infty, \quad \lim_{x\to 0^+} \frac{x-2}{x(x+7)} = -\infty.$$
When $k=2$, we find that
$$\lim_{x\to 2} \frac{x-2}{x(x+7)} = \frac{2-2}{2(2+7)} = 0.$$

2. Suppose that
$$\lim_{x\to a} \frac{\sqrt{x}-\sqrt{a}+\sqrt{x-a}}{\sqrt{x^2-a^2}} = \frac{1}{2}.$$
Determine the value of a.

Let $y=\sqrt{x}$, and $b=\sqrt{a}$. We compute the limit
$$\begin{aligned}
\lim_{y\to b} \frac{y-b+\sqrt{y^2-b^2}}{\sqrt{y^4-b^4}} &= \lim_{y\to b} \frac{y-b+\sqrt{y^2-b^2}}{\sqrt{y^4-b^4}} \\
&= \lim_{y\to b} \frac{\left(\sqrt{y-b}\right)^2+\left(\sqrt{y-b}\right)\left(\sqrt{y+b}\right)}{\sqrt{y^2+b^2}\sqrt{y^2-b^2}} \\
&= \lim_{y\to b} \frac{\left(\sqrt{y-b}\right)^2+\left(\sqrt{y-b}\right)\left(\sqrt{y+b}\right)}{\sqrt{y^2+b^2}\sqrt{y+b}\sqrt{y-b}} \\
&= \lim_{y\to b} \frac{\sqrt{y-b}+\sqrt{y+b}}{\sqrt{y^2+b^2}\sqrt{y+b}} = \frac{\sqrt{0}+\sqrt{2b}}{\sqrt{2b^2}\sqrt{2b}} = \frac{1}{\sqrt{2}b}.
\end{aligned}$$
We thus require that
$$\frac{1}{\sqrt{2}b} = \frac{1}{2} \quad \Longrightarrow \quad b = \frac{2}{\sqrt{2}} = \sqrt{2}.$$
Since $b=\sqrt{a}$, we find that $a=2$.

3. Compute the following limits at infinity:

(a) $\displaystyle\lim_{x\to\infty} \frac{2x+3}{5x+7}$

(b) $\displaystyle\lim_{x\to-\infty} \frac{x^2-4x+8}{3x^3}$

(c) $\displaystyle\lim_{x\to\infty} \frac{\cos x - 1}{x}$

(a) We divide the expression throughout by the highest power:

$$\lim_{x\to\infty} \frac{2x+3}{5x+7} \cdot \frac{1/x}{1/x} = \lim_{x\to\infty} \frac{2+\cancelto{0}{3/x}}{5+\cancelto{0}{7/x}} = \frac{2}{5}.$$

(b) Simplifying, we have

$$\lim_{x\to-\infty} \frac{x^2-4x+8}{3x^3} = \lim_{x\to-\infty} \cancelto{0}{\frac{1}{3x}} - \cancelto{0}{\frac{4}{3x^2}} + \cancelto{0}{\frac{8}{3x^3}} = 0.$$

(c) We can rewrite the limit as

$$\lim_{x\to\infty} \frac{\cos x}{x} - \frac{1}{x} = \lim_{x\to\infty} \frac{\cos x}{x} - \lim_{x\to\infty} \frac{1}{x}.$$

To evaluate the first limit, recall that the range of $\cos x$ is given by

$$-1 \le \cos x \le 1.$$

In particular, for all $x > 0$ we have

$$-\frac{1}{x} \le \frac{\cos x}{x} \le \frac{1}{x}.$$

We can thus apply the Squeeze Theorem: since both $-1/x$ and $1/x$ approach 0 as x gets larger and larger, we find that

$$0 \le \lim_{x\to\infty} \frac{\cos x}{x} \le 0 \quad\Longrightarrow\quad \lim_{x\to\infty} \frac{\cos x}{x} = 0.$$

Thus, we can conclude that

$$\lim_{x\to\infty} \frac{\cos x - 1}{x} = \lim_{x\to\infty} \frac{\cos x}{x} - \lim_{x\to\infty} \frac{1}{x} = 0 - 0 = 0.$$

4. Compute the derivatives of each of the following functions.

(a) $y = (x+1)^2 \left(x^2 + 2x\right)$

(b) $r = \left(\dfrac{1+\sin\theta}{1-\cos\theta}\right)^2$

(c) $y = \ln(\ln(\ln x))$

(a) We apply the product rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\left[(x+1)^2\right]\left(x^2+2x\right) + (x+1)^2 \frac{d}{dx}\left[x^2+2x\right] \\ &= 2(x+1)\left(x^2+2x\right) + (x+1)^2(2x+2) \\ &= 2(x+1)\left(x^2+2x+(x+1)^2\right).\end{aligned}$$

(b) We first apply the chain rule:

$$\frac{dr}{d\theta} = 2 \cdot \frac{1+\sin\theta}{1-\cos\theta} \cdot \frac{d}{d\theta}\left[\frac{1+\sin\theta}{1-\cos\theta}\right].$$

We then apply the quotient rule:

$$\begin{aligned}\frac{dr}{d\theta} &= 2 \cdot \frac{1+\sin\theta}{1-\cos\theta} \cdot \frac{(1-\cos\theta)\frac{d}{d\theta}[1+\sin\theta] - \frac{d}{d\theta}[1-\cos\theta](1+\sin\theta)}{(1-\cos\theta)^2} \\ &= 2 \cdot \frac{1+\sin\theta}{1-\cos\theta} \cdot \frac{(1-\cos\theta)\cos\theta - \sin\theta(1+\sin\theta)}{(1-\cos\theta)^2} \\ &= 2 \cdot \frac{1+\sin\theta}{1-\cos\theta} \cdot \frac{\cos\theta - \cos^2\theta - \sin\theta - \sin^2\theta}{(1-\cos\theta)^2} \\ &= 2(1+\sin\theta) \cdot \frac{\cos\theta - \sin\theta - 1}{(1-\cos\theta)^3}.\end{aligned}$$

(c) We repeatedly apply the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{\ln(\ln x)} \frac{d}{dx}[\ln(\ln x)] \\ &= \frac{1}{\ln(\ln x)} \cdot \frac{1}{\ln x} \cdot \frac{d}{dx}[\ln x] \\ &= \frac{1}{\ln(\ln x)} \cdot \frac{1}{\ln x} \cdot \frac{1}{x} = \frac{1}{x \cdot \ln x \cdot \ln(\ln x)}.\end{aligned}$$

5. Determine dy/dx.

   (a) $\ln(x/y) = 1$

   (b) $x^2 - y^2 = 1$

   (c) $x^y = \sqrt{2}$

(a) By the laws of logarithms, we may rewrite the equation as

$$\ln x - \ln y = 1 \quad \Longrightarrow \quad \ln y = \ln x - 1 \quad \Longrightarrow \quad y = e^{\ln x - 1} = xe^{-1}.$$

Differentiating both sides of the equation, we find that

$$\frac{dy}{dx} = \frac{1}{e}.$$

Alternatively, we can differentiate both sides of the equation with respect to x, evaluating the derivative of the left-hand side implicitly:

$$\frac{1}{x/y} \cdot \frac{\frac{d}{dx}[x]\, y - x\frac{dy}{dx}}{y^2} = 0 \quad \Longrightarrow \quad \frac{y}{x} \cdot \frac{y - x\frac{dy}{dx}}{y^2} = 0.$$

We simplify the above expression and find dy/dx:

$$\frac{y - x\frac{dy}{dx}}{xy} = 0 \quad \Longrightarrow \quad y - x\frac{dy}{dx} = 0 \quad \Longrightarrow \quad \frac{dy}{dx} = \frac{y}{x}.$$

Note that, since $\ln(x/y) = 1$, we have $x/y = e$, which implies that

$$\frac{dy}{dx} = \frac{y}{x} = \frac{1}{e}.$$

(b) We differentiate both sides of the equation with respect to x, evaluating the derivative of the left-hand side implicitly:

$$2x - 2y\frac{dy}{dx} = 0 \quad \Longrightarrow \quad 2x = 2y\frac{dy}{dx} \quad \Longrightarrow \quad \frac{dy}{dx} = \frac{x}{y}.$$

(c) We take the logarithm of both sides of the equation:

$$\ln x^y = \ln\sqrt{2} \quad \Longrightarrow \quad y \ln x = \ln\sqrt{2}.$$

We proceed to differentiate both sides with respect to x, evaluating the derivative of the left-hand side implicitly:

$$\frac{dy}{dx} \cdot \ln x + y \cdot \frac{1}{x} = 0 \quad \Longrightarrow \quad \frac{dy}{dx} = -\frac{y}{x \ln x}.$$

6. Consider the curve given by $y = 1 + \cos x$. Find the equations of the tangent line and normal line to the curve when $x = \pi/2$. Sketch the graph of the curve with the tangent and normal lines at the given point.

We first determine the tangent line to the graph of $y = 1 + \cos x$ at $x = \pi/2$. To begin, note that when $x = \pi/2$,

$$y = 1 + \cos\frac{\pi}{2} = 1 + 0 = 1,$$

and so the line must pass through the point $(\pi/2, 1)$. The slope m of the tangent line is given by the derivative:

$$\frac{dy}{dx} = \frac{d}{dx}[1 + \cos x] = -\sin x \quad \Longrightarrow \quad m = \left.\frac{dy}{dx}\right|_{x=\pi/2} = -\sin\frac{\pi}{2} = -1.$$

Hence, the equation of the tangent line is given by

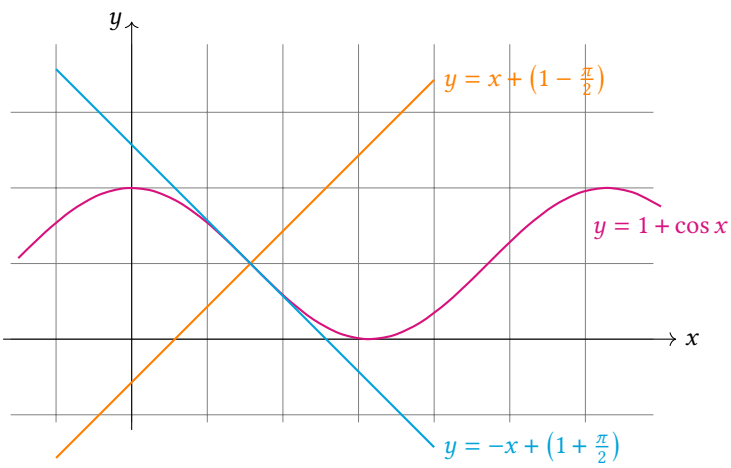
$$y = -\left(x - \frac{\pi}{2}\right) + 1 \quad \Longrightarrow \quad y = -x + \left(1 + \frac{\pi}{2}\right).$$

Recall that the normal line must be perpendicular to the tangent line at the point $(\pi/2, 1)$: thus, its slope must be

$$-\frac{1}{m} = -\frac{1}{-1} = 1,$$

and its equation is thus

$$y = x - \frac{\pi}{2} + 1 \quad \Longrightarrow \quad y = x + \left(1 - \frac{\pi}{2}\right).$$



7. An ant traverses the xy-plane with a horizontal speed of $dx/dt = 10$ mm sec^{-1} and a vertical speed of $dy/dt = 5$ mm sec^{-1}. How fast is the ant travelling away from the origin as it passes through the point $(3, -4)$?

   The ant's distance from the origin is given by the function

$$f(x, y) = \sqrt{x^2 + y^2}.$$

   To determine the rate of change in the ants distance with respect to time, we find the derivative df/dt:

$$\frac{df}{dt} = \frac{1}{2}\left(x^2 + y^2\right)^{-1/2}\left(2x\frac{dx}{dt} + 2y\frac{dy}{dt}\right).$$

   Since the ant is at $x = 3$ and $y = -4$, and we have the values of dx/dt and dy/dt, we can compute

$$\begin{aligned}\left.\frac{df}{dt}\right|_{(3,-4)} &= \frac{1}{2\sqrt{3^2 + (-4)^2}}\left(2 \cdot 3 \cdot 10 + 2 \cdot -4 \cdot 5\right) \\ &= \frac{1}{2 \cdot 5}\left(60 - 40\right) = \frac{20}{10} = 2.\end{aligned}$$

   Hence, the ant is traveling at a speed of 2 mm sec^{-1} away from the origin when it is at the point $(3, -4)$.

8. Sketch the graphs of the following functions by hand.

(a) $y = x^4 - 2x^2$

(b) $y = e^{-2x^2}$

(c) $y = \dfrac{x+1}{x-3}$

(a) To begin, we find the points where the graph cuts the axes.

- When $x = 0$, we have $y = 0$.
- When $y = 0$, we have $x^2\left(x^2 - 2\right) = 0$, and so $x = 0, \pm\sqrt{2}$.

Thus, the graph must pass through $(0,0)$, $\left(\sqrt{2}, 0\right)$, and $\left(-\sqrt{2}, 0\right)$.
We now determine the stationary points:

$$\frac{dy}{dx} = 4x^3 - 4x = 0 \quad \Longrightarrow \quad 4x\left(x^2 - 1\right) = 4x\,(x+1)\,(x-1) = 0.$$

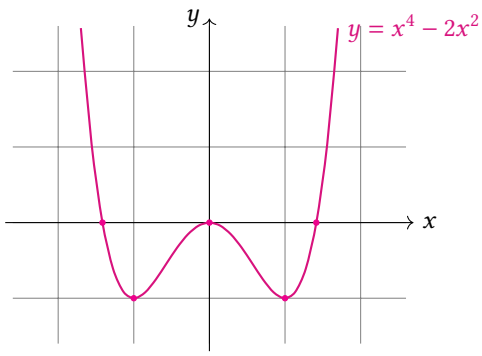
There are thus three stationary points: at

- $x = 0$, with $y = 0 - 0 = 0$,
- $x = -1$, with $y = (-1)^4 - 2\,(-1)^2 = -1$, and
- $x = 1$, with $y = 1^4 - 2 \cdot 1^2 = -1$.

We consider the sign of dy/dx outside the stationary points:

Interval	$4x$	$x+1$	$x-1$	$y'(x)$	$y(x)$
$(-\infty, -1)$	$-$	$-$	$-$	$-$	$\searrow$
$(-1, 0)$	$-$	$+$	$-$	$+$	$\nearrow$
$(0, 1)$	$+$	$+$	$-$	$-$	$\searrow$
$(1, \infty)$	$+$	$+$	$+$	$+$	$\nearrow$

We now sketch the graph of $y = x^4 - 2x^2$, noting down the stationary points and the intercepts.



For a more accurate sketch, we find the inflection points of the curve by examining when the second derivative is 0:

$$\frac{d^2y}{dx^2} = 12x^2 - 4 = 4\left(3x^2 - 1\right) = 0 \quad \Longrightarrow \quad x = \pm\frac{1}{\sqrt{3}}.$$

We consider the sign of d^2y/dx^2 outside of the inflection points:

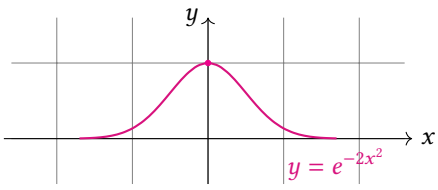
Interval	$y''(x)$	Concavity of $f(x)$
$\left(-\infty, -1/\sqrt{3}\right)$	$+$	$\smile$
$\left(-1/\sqrt{3}, 1/\sqrt{3}\right)$	$-$	$\frown$
$\left(1/\sqrt{3}, \infty\right)$	$+$	$\smile$

(b) Observe that $y = e^{-2x^2}$ is always positive and must lie above the x-axis; it has an y-intercept at $(0, 1)$. The stationary points occur at:

$$\frac{dy}{dx} = e^{-2x^2} \cdot (-4x) = -4xe^{-2x^2} = 0 \quad \Longrightarrow \quad x = 0.$$

This is precisely the y-intercept. Outside of this point, we find that

- when $x < 0$, $dy/dx > 0$ and y is increasing,
- when $x > 0$, $dy/dx < 0$ and y is decreasing.



We can check the second derivative to determine the inflection points:

$$\frac{d^2y}{dx^2} = -4e^{-2x^2} - 4xe^{-2x^2}(-4x) = \left(16x^2 - 4\right)e^{-2x^2} = 0 \implies x = \pm\frac{1}{2}.$$

We consider the sign of d^2y/dx^2 outside of the inflection points:

Interval	$y''(x)$	Concavity of $f(x)$
$(-\infty, -1/2)$	$+$	$\smile$
$(-1/2, 1/2)$	$-$	$\frown$
$(1/2, \infty)$	$+$	$\smile$

(c) To begin, observe that

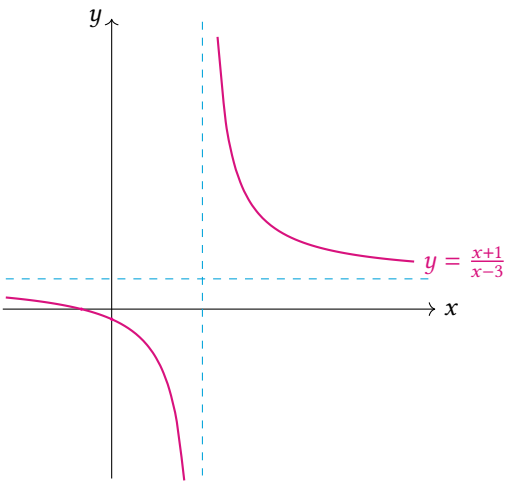
$$y = \frac{x+1}{x-3} = \frac{x-3+4}{x-3} = 1 + \frac{4}{x-3}.$$

Thus, the graph has a vertical asymptote at $x - 3 = 0$, or $x = 3$, and a horizontal asymptote at $y = 1$. We determine the intercepts of the graph: when $x = 0$, $y = -1/3$, and when $y = 0$, $x = -1$; so the graph cuts the axes at $(0, -1/3)$ and $(-1, 0)$.

We now determine the stationary points:

$$\frac{dy}{dx} = \frac{(x-3)-(x+1)}{(x-3)^2} = -\frac{4}{(x-3)^2} \neq 0.$$

The graph thus has no stationary points; moreover, $dy/dx < 0$ for all x, and so the function must always be decreasing.



9. Find the dimensions of the rectangle of perimeter 100 m with the largest possible area.

Let x be the length of the rectangle, and let y be its width. Since its perimeter is 100, we have

$$2x + 2y = 100 \quad \Longrightarrow \quad y = 50 - x.$$

Thus, the area of the rectangle is given by

$$A(x) = xy = x(50 - x) = -x^2 + 50x.$$

We wish to find the value of x for which A is a maximum: setting its derivative to 0, we have

$$A'(x) = -2x + 50 = 0 \quad \Longrightarrow \quad x = 25.$$

Observe that, indeed, this is a maximum, since $A(x) > 0$ when $x < 25$ and $A(x) < 0$ when $x > 25$. Thus, $x = 25$ and $y = 50 - 25 = 25$, so the dimensions of the rectangle with the largest area must be 25×25.

10. Find the area of the largest possible rectangle that can be inscribed in the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Since the rectangle is centered at the origin, it must have length $2x$ and width $2y$, such that its area is given by

$$A = (2x)(2y) = 4xy.$$

Treating y as a function of x, we have

$$\frac{dA}{dx} = 4y + 4x\frac{dy}{dx}.$$

We now implicitly differentiate the equation of the ellipse to determine dy/dx:

$$\frac{1}{a^2} \cdot 2x + \frac{1}{b^2} 2y \frac{dy}{dx} = 0 \quad \Longrightarrow \quad \frac{dy}{dx} = -\frac{2x}{a^2} \cdot \frac{b^2}{2y} = -\frac{b^2 x}{a^2 y}.$$

Hence, we have

$$\frac{dA}{dx} = 4y + 4x\left(-\frac{b^2 x}{a^2 y}\right) = 4y - 4\frac{b^2 x^2}{a^2 y}.$$

We determine the maximum of $A(x)$ by setting its derivative to 0:

$$4y - 4\frac{b^2 x^2}{a^2 y} = 0 \quad \Longrightarrow \quad y = \frac{b^2 x^2}{a^2 y} \quad \Longrightarrow \quad \frac{x^2}{a^2} = \frac{y^2}{b^2}.$$

Plugging this into the ellipse equation, we find that

$$\frac{x^2}{a^2} + \frac{x^2}{a^2} = 1 \quad \Longrightarrow \quad x = \frac{a}{\sqrt{2}}.$$

Similarly, we have

$$\frac{y^2}{b^2} + \frac{y^2}{b^2} = 1 \quad \Longrightarrow \quad y = \frac{b}{\sqrt{2}}.$$

Thus, the area of the largest possible rectangle is given by

$$A = 4xy = 4 \cdot \frac{a}{\sqrt{2}} \cdot \frac{b}{\sqrt{2}} = 2ab.$$

11. Use l'Hôpital's rule to evaluate the following limits:

(a) $\lim_{x\to 0} \dfrac{\tan x}{x+\sin x}$

(b) $\lim_{x\to 0} \dfrac{3^x-1}{x}$

(c) $\lim_{x\to\infty} \sqrt{x^2+x+1}-\sqrt{x^2-x}$

(a) Observe that

$$\frac{\tan 0}{0+\sin 0} = \frac{0}{0},$$

and we may thus use l'Hôpital's rule to evaluate the limit.

$$\lim_{x\to 0} \frac{\tan x}{x+\sin x} = \lim_{x\to 0} \frac{\frac{d}{dx}\tan x}{\frac{d}{dx}[x+\sin x]} = \lim_{x\to 0} \frac{\sec^2 x}{1+\cos x} = \frac{1}{1+1} = \frac{1}{2}.$$

(b) Observe that

$$\frac{3^0-1}{0} = \frac{0}{0},$$

and we can use l'Hôpital's rule. To compute the derivative of 3^x, let

$$y = 3^x \quad\Longrightarrow\quad \ln y = x\ln 3.$$

We perform implicit differentiation:

$$\frac{1}{y}\frac{dy}{dx} = \ln 3 \quad\Longrightarrow\quad \frac{dy}{dx} = y\ln 3 = 3^x\ln 3.$$

Thus,

$$\lim_{x\to 0} \frac{3^x-1}{x} = \lim_{x\to 0} \frac{\frac{d}{dx}[3^x-1]}{\frac{d}{dx}x} = \lim_{x\to 0} \frac{3^x\ln 3}{1} = \ln 3.$$

(c) We begin by rationalizing the expression:

$$\left(\sqrt{x^2+x+1}-\sqrt{x^2-x}\right)\frac{\sqrt{x^2+x+1}+\sqrt{x^2-x}}{\sqrt{x^2+x+1}+\sqrt{x^2-x}} = \frac{2x+1}{\sqrt{x^2+x+1}+\sqrt{x^2-x}}.$$

Thus, we consider the limit

$$\lim_{x\to\infty} \frac{2x+1}{\sqrt{x^2+x+1}+\sqrt{x^2-x}} = \frac{\infty}{\infty}.$$

We apply l'Hôpital's rule. The derivative of the numerator is simply

$$\frac{d}{dx}[2x+1] = 2.$$

The derivative of the denominator is given by

$$\frac{d}{dx}\sqrt{x^2+x+1}+\sqrt{x^2-x} = \frac{1}{2}\left(\frac{2x+1}{\sqrt{x^2+x+1}}+\frac{2x-1}{\sqrt{x^2-x}}\right).$$

To compute the limit to infinity, observe that

$$\frac{1}{2}\left(\frac{2x+1}{\sqrt{x^2+x+1}}+\frac{2x-1}{\sqrt{x^2-x}}\right)\cdot\frac{1/x}{1/x} = \frac{1}{2}\left(\frac{2+\frac{1}{x}}{\sqrt{1+\frac{1}{x}+\frac{1}{x^2}}}+\frac{2-\frac{1}{x}}{\sqrt{1-\frac{1}{x}}}\right).$$

Note that the fractions $1/x$ and $1/x^2$ approach 0 as x gets large, and so the above expression approaches $4/2 = 2$. We can finally use l'Hôpital's rule to evaluate the original limit:

$$\lim_{x\to\infty} \sqrt{x^2+x+1}-\sqrt{x^2-x} = \lim_{x\to\infty} \frac{2x+1}{\sqrt{x^2+x+1}+\sqrt{x^2-x}} = \frac{2}{2} = 1.$$

12. When a violin string is plucked and released, it vibrates at a frequency given by

$$f = \frac{1}{2L}\sqrt{\frac{T}{\rho}},$$

where L is the length of the string, T is its tension, and ρ is its linear density.

(a) Find the rate at which frequency changes with respect to

i. length (when T and ρ are constant);

ii. tension (when L and ρ are constant); and

iii. linear density (when L and T are constant).

(b) The value of f determines the pitch of a note, that is, how high or low the note sounds. (The higher the frequency, the higher the pitch.) Based on your answers to the previous part, determine what happens to the pitch of a note when:

i. the effective length of a string is decreased by placing a finger on the string so a shorter portion of the string vibrates;

ii. the tension is increased by turning a tuning peg;

iii. the linear density is increased by switching to another string.

(a) We compute the derivatives of f with respect to L, T, and ρ respectively:

$$\frac{df}{dL} = -\frac{1}{2L^2}\sqrt{\frac{T}{\rho}},$$

$$\frac{df}{dT} = \frac{1}{2L}\cdot\frac{1}{2}\left(\frac{T}{\rho}\right)^{-1/2}\cdot\frac{1}{\rho} = \frac{1}{4L\sqrt{T\rho}},$$

$$\frac{df}{d\rho} = \frac{1}{2L}\cdot\frac{1}{2}\left(\frac{T}{\rho}\right)^{-1/2}\cdot -\frac{T}{\rho^2} = -\frac{1}{4L}\sqrt{\frac{T}{\rho^3}}.$$

(b) i. Observe that $df/dL < 0$: thus, a decrease in L will correspond to an increase in f, and the pitch must increase as well.

ii. Since $df/dT > 0$, an increase in T will correspond to an increase in f, and the pitch increases.

iii. Since $df/d\rho < 0$, an increase in ρ will correspond to a decrease in f, and so the pitch decreases.